XAI Foundations & Interpretable Models

Lecture 1 – Explainable AI

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Week 01

opencampus.sh

Agenda

Why explainability matters

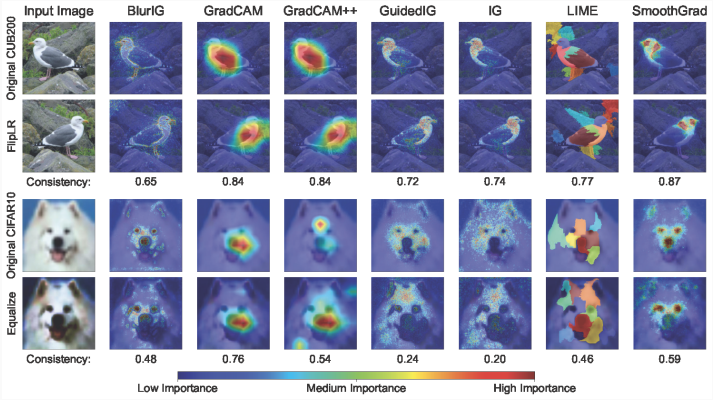
Taxonomy

Linear and logistic regression

Decision trees

Summary

What do explanations look like? Vision



Saliency maps show which pixels support a prediction.

- Methods give different maps for the same image
- Harmless changes (flip, equalization) can change the map
- Consistency between the two maps: 0.87 down to 0.20

Rows: original vs. transformed input (CUB200, CIFAR10); columns: attribution methods. See Lectures 6 and 8.

A model can be right for the wrong reason

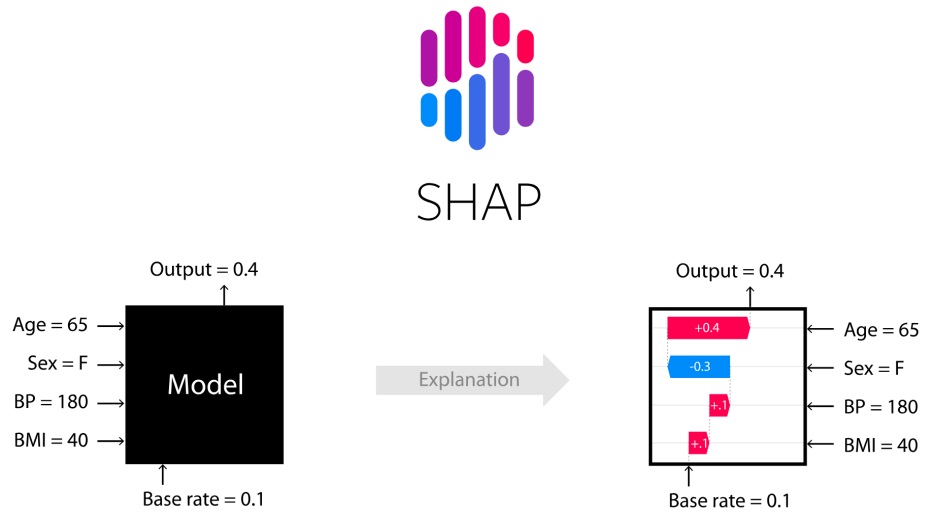
Classic cautionary examples

- Husky vs. wolf: a classifier keyed on snow in the background (Ribeiro et al., 2016)
- Pneumonia risk: a model learned that asthma lowers risk, because asthma patients received intensive care (Caruana et al., 2015)

Lesson

High test accuracy does not show that the model learned the intended concept. Explanations help us debug shortcuts and confounders before deployment.

What do explanations look like? Tabular data (SHAP)



Why do we need explanations?

Debugging

- Find shortcuts and data leakage
- Improve features and models

Trust & fairness

- Which inputs drive a decision?
- Are protected attributes or proxies involved?

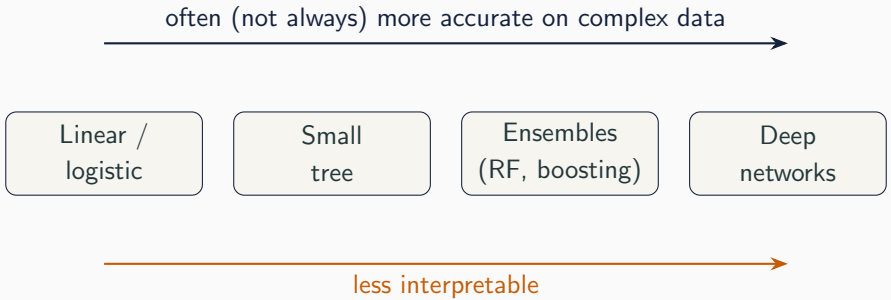
Regulation

- GDPR rules on automated decisions
- EU AI Act: transparency and human oversight for high-risk systems

Who is the explanation for?

Developers, domain experts, affected individuals and auditors need *different* explanations.

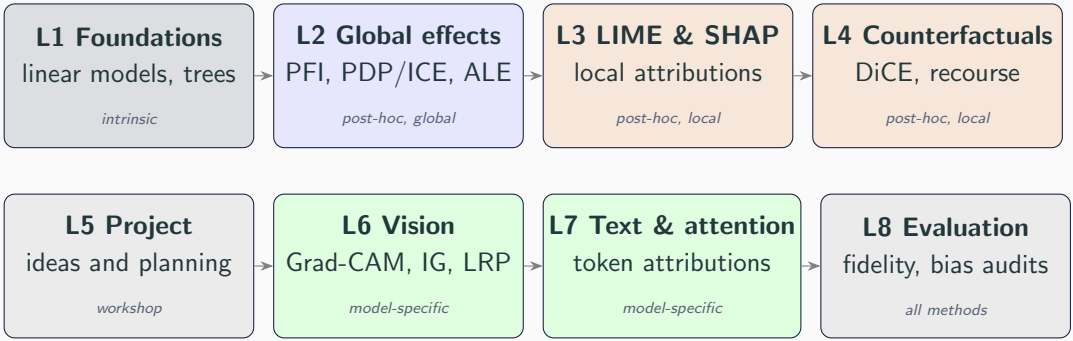
Interpretability–accuracy tradeoff



Practical rule

On tabular data, simple models are often competitive. Always fit an interpretable baseline first.

Course map


L1 Foundations
linear models, trees
intrinsic
L2 Global effects
PFI, PDP/ICE, ALE
post-hoc, global
L3 LIME & SHAP
local attributions
post-hoc, local
L4 Counterfactuals
DiCE, recourse
post-hoc, local
L5 Project
ideas and planning
workshop
L6 Vision
Grad-CAM, IG, LRP
model-specific
L7 Text & attention
token attributions
model-specific
L8 Evaluation
fidelity, bias audits
all methods

Three questions to classify any XAI method

1. When?

Built in, or added afterwards?

Intrinsic: the model *is* the explanation

linear weights, tree rules

Post-hoc: explain a trained black box

SHAP, LIME, PDP

2. How much?

Whole model, or one prediction?

Global: overall model behavior

permutation importance

Local: why this prediction?

counterfactuals, LIME

3. What access?

Needs the model's internals?

Model-specific: uses weights or gradients

Grad-CAM, TreeSHAP

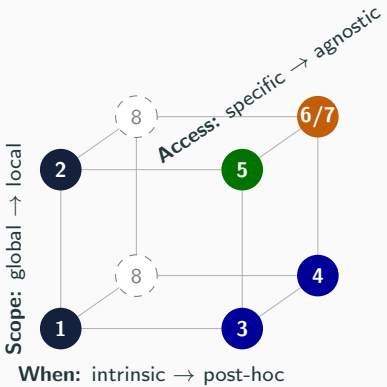
Model-agnostic: only inputs and outputs

PFI, KernelSHAP

Why it matters

The axes tell you which method *fits your question*: "Why was this loan rejected?" needs a local explanation; auditing a model needs a global one.

Three axes: where methods sit



1. Intrinsic, global, specific: linear weights, tree rules (L1)
2. Intrinsic, local, specific: tree decision path, $\beta_j x_j$ (L1)
3. Post-hoc, global, specific: mean $|\mathrm{TreeSHAP}|$ (L3)
4. Post-hoc, global, agnostic: PFI, PDP/ICE, ALE (L2)
5. Post-hoc, local, specific: Grad-CAM, IG, LRP, attention (L6–7)
6. Post-hoc, local, agnostic: LIME, KernelSHAP (L3)
7. Post-hoc, local, agnostic: counterfactuals, DiCE (L4)
8. Intrinsic + agnostic: not applicable (an interpretable-by-design model is tied to its own structure)

Simplified: some methods (e.g. SHAP) occupy several corners.

Reading linear models

Linear regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

β_j: change in $\hat{y}$ for a one-unit increase in x_j, all other features fixed.

Logistic regression

$$\log \frac{P(y = 1 \mid x)}{1 - P(y = 1 \mid x)} = \beta_0 + \sum_j \beta_j x_j \qquad \Rightarrow \qquad \text{odds ratio} = e^{\beta_j}$$

Pitfalls when interpreting coefficients

- Scale: compare coefficients only after standardizing features
- Correlated features: coefficients become unstable and can flip sign
- Linearity: the effect of x_j is assumed constant across its range
- Causality: coefficients describe the model, not the world

Code: odds ratios in scikit-learn

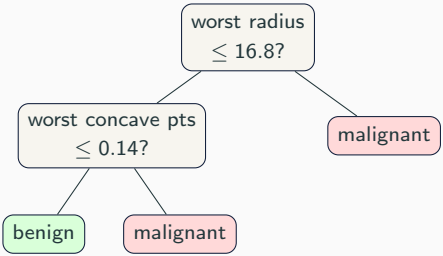
```
from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

X, y = load_breast_cancer(return_X_y=True, as_frame=True)
Xs = StandardScaler().fit_transform(X)
model = LogisticRegression(max_iter=1000).fit(Xs, y)

odds = np.exp(model.coef_[0]) # per 1 std. dev.
pd.Series(odds, index=X.columns).sort_values()
```

Note: in this dataset target 1 means *benign*.

Decision trees: predictions as rules



Schematic example, not fitted values.

- Each prediction is a path of if–then rules
- Handles non-linearity and interactions
- Depth controls the tradeoff: shallow = readable, deep = accurate but opaque

Code: depth vs. accuracy and tree rules

```
from sklearn.tree import DecisionTreeClassifier, export_text

for d in range(1, 16):
    t = DecisionTreeClassifier(max_depth=d, random_state=0).fit(X_train, y_train)
    print(d, t.score(X_train, y_train), t.score(X_test, y_test))

small = DecisionTreeClassifier(max_depth=3, random_state=0).fit(X_train, y_train)
print(export_text(small, feature_names=list(X.columns)))
```

Limitations of trees

- Instability: small changes in the training data can change the tree substantially
- Axis-aligned splits: smooth or diagonal boundaries need many splits
- Deep trees are not interpretable: 100 leaves is not an explanation

Notebook task: train the same tree on bootstrap samples and compare the root splits.

Summary and hands-on

Key takeaways

- Explanations are for debugging, trust, fairness and compliance
- Taxonomy: intrinsic vs. post-hoc, global vs. local, specific vs. agnostic
- Linear models and small trees are interpretable, with caveats

Hands-on notebook

1. Odds ratios
2. Accuracy vs. depth
3. Tree rules
4. Bootstrap stability

References

- Molnar, C. *Interpretable Machine Learning* (online book)
- Ribeiro, Singh, Guestrin (2016). "Why Should I Trust You?" KDD
- Caruana et al. (2015). Intelligible models for healthcare. KDD
- Rudin (2019). Stop explaining black box models for high-stakes decisions. *Nature Machine Intelligence*